PROJECT DOCUMENTATION

AI StudyBuddy - AI Based Smart Learning Platform

INTRODUCTION
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StudyBuddy is an AI-powered learning platform developed to make studying easier, faster, and more effective for students. It uses Generative AI to help students work with their study materials and create useful learning resources such as summaries, flashcards, quizzes, and personalized study plans. By bringing these learning activities into one platform, StudyBuddy helps students save time, understand their subjects more easily, revise important concepts, and check their knowledge through practice. The main purpose of StudyBuddy is to provide students with simple and intelligent support throughout their learning process.

Slide 2 – Project Overview

StudyBuddy is a smart educational application designed to support students during their learning and examination preparation.

The platform uses Artificial Intelligence to understand learning content and provide useful study resources.

StudyBuddy helps students to:

- Understand difficult topics
- Revise lessons quickly
- Practice questions
- Organize their study time
- Keep learning resources in one place

Slide 3 – Existing Learning Challenges

Students commonly face problems such as:

- Spending too much time reading lengthy materials
- Difficulty identifying important concepts
- Preparing notes manually
- Forgetting previously studied topics
- Lack of regular revision
- Difficulty creating a proper study timetable
- Searching different websites and applications for study resources

These difficulties can make examination preparation stressful and time-consuming.

Slide 4 – Our Proposed Solution

StudyBuddy provides an **AI-supported learning environment** where students can use their own study material and receive personalized learning assistance.

The platform provides:

- Input Study Material
- AI Understanding
- Learning Content Creation
- Student Practice
- Revision & Progress

Slide 5 – Project Goals

The major goals of StudyBuddy are:

- Make learning easier and more organized
- Reduce the time required for revision
- Provide personalized learning assistance
- Convert study material into useful resources
- Encourage regular practice
- Help students prepare for examinations
- Provide a single platform for different study activities

Slide 6 – Major Modules

StudyBuddy can be divided into the following modules:

User Module

Registration, login and profile management.

Study Material Module

Adding and organizing learning content.

AI Learning Module

Generating useful learning resources.

Practice Module

Quizzes and revision questions.

Planning Module

Creating personalized study schedules.

Admin Module

Managing users and monitoring the platform.

Slide 7 – AI Learning Features

The AI component makes StudyBuddy more interactive.

Smart Summary

Converts lengthy content into shorter revision notes.

Question Generator

Creates questions from the student's study material.

Flashcard Generator

Produces question-and-answer cards for quick practice.

Quiz Generator

Creates multiple-choice questions to test understanding.

Study Planner

Suggests a learning schedule according to the student's requirements.

Slide 8 – Student Experience

A student can use StudyBuddy in a simple sequence:

Create Account

↓

Login

↓

Add Study Material

↓

Choose Learning Option

↓

AI Processes Content

↓

Receive Learning Resources

↓

Practice & Revise

↓

Save Resources

This makes the complete learning process easier to manage.

Slide 9 – System Components

The application consists of several major components:

- **Frontend** – Student interaction
- **Backend Server** – Application processing
- **Authentication Layer** – User verification
- **AI Service** – Intelligent content generation
- **Database** – Storage of user and learning information
- **API Layer** – Communication between different components

Slide 10 – Technology Used

Frontend

React.js

Backend

Node.js
Express.js

Database

MongoDB
Mongoose

AI Technology

Google Gemini API

Security

- JWT Authentication
- bcrypt

Development Tools

VisualStudioCode
Postman
npm

Slide 11 – Application Architecture

StudyBuddy Architecture

Student

↓

React Application

↓

REST API

↓

Node.js + Express.js

↓

Application Services

↙ ↘

MongoDB **AI Service**

↓

Google Gemini

The architecture allows each part of the application to perform a specific responsibility.

Slide 12 – Data Management

StudyBuddy uses MongoDB to maintain application data.

Important Data

- User information
- Study subjects
- Learning materials

- Generated summaries
- Flashcards
- Quiz questions
- Study schedules

Mongoose is used to define the structure of data and communicate between the Node.js application and MongoDB.

Slide 13 – Security Features

Protecting student information is an important part of the system.

Authentication

Users must log in before accessing protected features.

JWT

Tokens are used to identify authenticated users.

Password Protection

Passwords are securely hashed before being stored.

Access Control

Different permissions can be provided to students and administrators.

Slide 14 – Example Scenario

Student: Rahul

Rahul has an examination in two weeks.

He has several chapters to study but does not have enough time to prepare notes manually.

Using StudyBuddy:

Rahul uploads his study material

↓

StudyBuddy analyzes the content

↓

AI creates a short summary

↓

AI generates flashcards

↓

AI creates practice questions

↓

Rahul uses the generated resources for revision

This helps Rahul organize his preparation more effectively.

Slide 15 – Benefits

StudyBuddy provides several benefits:

- Faster revision
- Organized learning materials
- Reduced manual preparation
- Interactive practice
- Personalized study planning
- Easy access to generated resources
- Better use of study time
- Centralized learning environment

Slide 16 – Future Scope

The system can be expanded with additional features such as:

- Voice-based learning assistant
- AI doubt clarification
- Student performance analysis
- Progress dashboard

- Mobile application
- Multiple-language learning
- Daily study reminders
- Difficulty-based quizzes
- AI-based learning recommendations

Slide 17 – Expected Result

The expected result of StudyBuddy is to provide students with a convenient digital learning assistant.

The system can help students:

Learn → Practice → Revise → Track → Improve

Instead of using several separate applications, students can access multiple learning-support features through one platform.

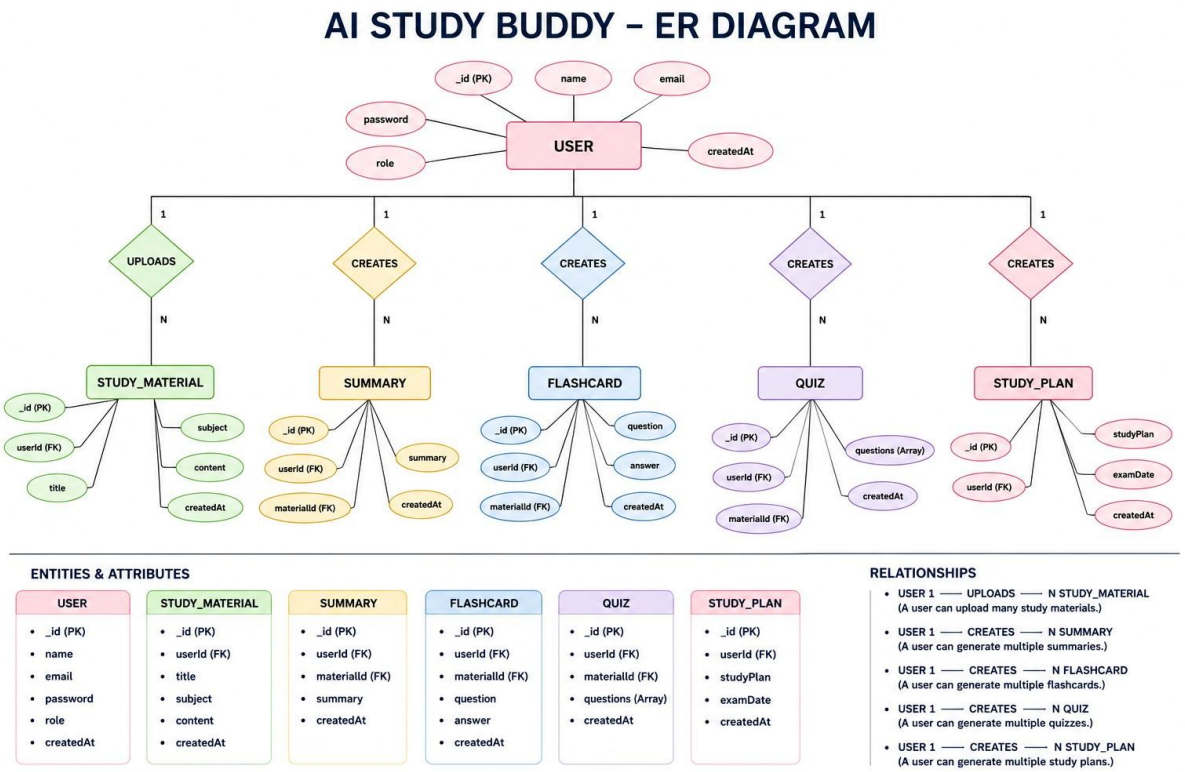
Slide 18 – Conclusion

StudyBuddy combines **Artificial Intelligence, web technologies, and database management** to create a smart learning environment.

The platform is designed to reduce repetitive study preparation and provide students with personalized resources.

By supporting summaries, flashcards, quizzes, and study planning, StudyBuddy can make the learning process more **organized, interactive, and convenient**.

Entity-Relationship (ER) Diagram Description



The AI StudyBuddy backend manages data through interconnected MongoDB collections that store user information, study materials, and AI-generated learning resources. Each entity is linked using logical relationships to ensure efficient data management and retrieval. **1. User Entity**

Stores the details of registered users and administrators.

Attributes:

- _id (Primary Key - ObjectId)
- name (String, Required)
- email (String, Required, Unique)

- password (String, Hashed)
- role (String - Student/Admin)
- createdAt

2. Study Material Entity

Stores study content uploaded by students for AI processing.

Attributes:

- _id (Primary Key)
- userId (Foreign Key → User)
- title
- subject
- content
- createdAt

3. Summary Entity

Stores AI-generated summaries.

Attributes:

- _id (Primary Key)
- userId (Foreign Key → User)
- materialId (Foreign Key → Study Material)
- summary
- createdAt

4. Flashcard Entity

Stores AI-generated flashcards.

Attributes:

- _id (Primary Key)
- userId (Foreign Key → User)
- materialId (Foreign Key → Study Material)
- question
- answer
- createdAt

-

5. Quiz Entity

Stores AI-generated quizzes.

Attributes:

- _id (Primary Key)
  userId (Foreign Key → User)
- materialId (Foreign Key → Study Material)
- questions ● createdAt

6. Study Plan Entity

Stores personalized study plans generated by AI.

Attributes:

- _id
  (Primary
  Key)
- userId
  (Foreign Key
  → User)
- studyPlan
- examDate
- createdAt

Key Relationships

- **User → Study Material (One-to-Many):** A user can upload multiple study materials.
- **Study Material → Summary (One-to-One):** Each uploaded material can have one AI-generated summary.
- **Study Material → Flashcards (One-to-Many):** One study material can generate multiple flashcards.
- **Study Material → Quiz (One-to-One):** One study material can generate one quiz containing multiple questions.
- **User → Study Plan (One-to-Many):** A user can generate multiple study plans based on different subjects or examination schedules.
- **User → AI Resources (One-to-Many):** All AI-generated summaries, flashcards, quizzes, and study plans are associated with the user who generated them.

Key Features

The AI StudyBuddy backend provides intelligent learning assistance through secure and scalable REST APIs. The major features of the system are as follows:

-
- **Secure User Authentication:** Implements JWT-based authentication with encrypted passwords using bcrypt for secure user access.
- **Study Material Management:** Allows users to upload and manage study materials that serve as input for AI-powered learning features.
- **AI Summary Generation:** Generates concise and easy-to-understand summaries from uploaded study content using the Google Gemini API.
- **AI Flashcard Generation:** Automatically creates flashcards from study materials to improve concept revision and active recall.
- **AI Quiz Generation:** Produces multiple-choice quizzes based on uploaded content, enabling students to test their knowledge.
  AI Study Plan Generation: Generates personalized study plans according to the user's learning goals and available study time.
- **Role-Based Access Control (RBAC):** Provides different access privileges for Students and Administrators to ensure secure system management.
- **MongoDB Data Storage:** Efficiently stores user information, uploaded materials, and AI-generated learning resources using MongoDB.

Roles and Responsibilities

Student

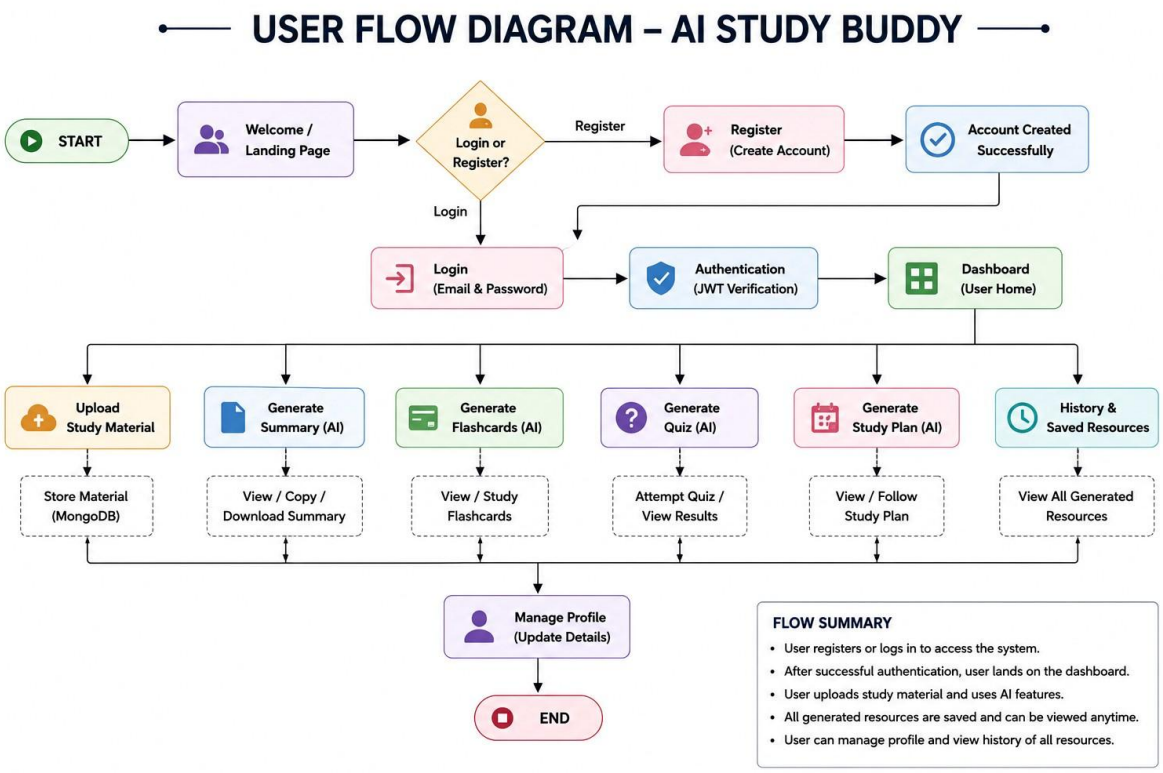
Students are the primary users of the system. They can register, log in securely, upload study materials, and utilize AI-powered features such as summary generation, flashcard creation, quiz generation, and personalized study plans. Students can also view and manage their previously generated learning resources.

Administrator

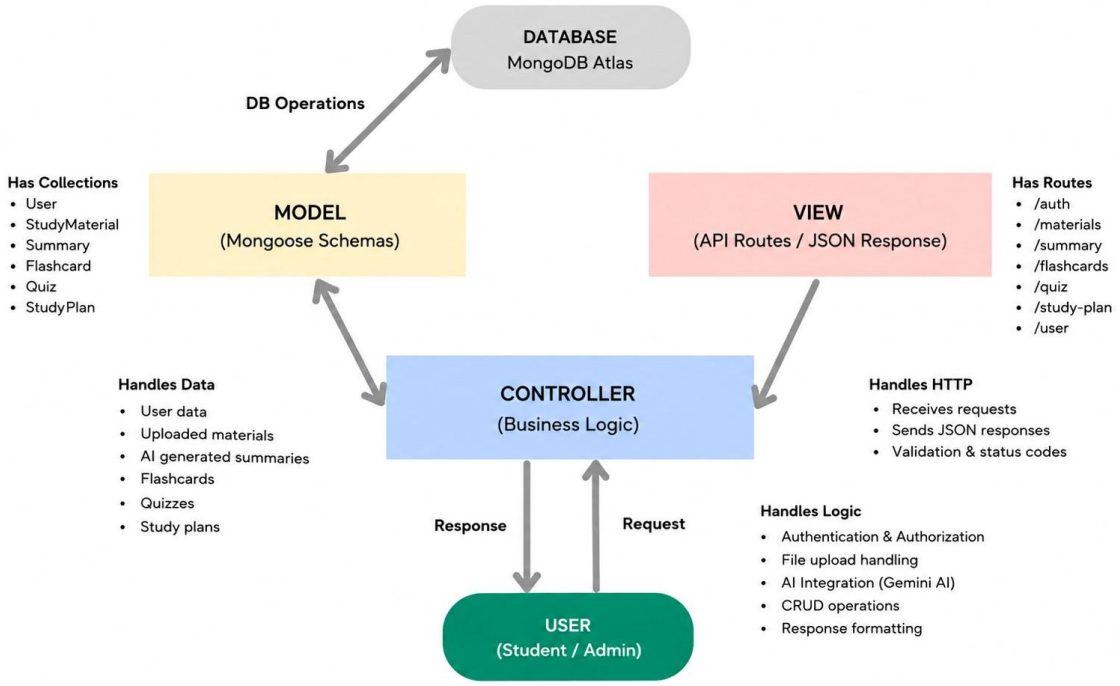
Administrators manage the overall system and ensure its smooth operation. They monitor registered users, manage user accounts, oversee application usage, maintain system integrity, and ensure that AI-powered services function correctly. Administrators also have permission to perform system-level management tasks and resolve user-related issues.

User Flow

The AI StudyBuddy application begins with the user registering or logging into the system. After successful JWT-based authentication, the user is redirected to the dashboard, where they can upload study materials and access AI-powered features. Based on the uploaded content, the system generates summaries, flashcards, quizzes, or personalized study plans using the Google Gemini API. All generated resources are displayed to the user and can be saved for future reference. Users can also manage their profile and view their previously generated learning resources through the dashboard.



MVC Architecture Pattern



The AI StudyBuddy backend follows the **Model-View-Controller (MVC)** architecture to organize the application into separate layers, making the code modular, maintainable, and scalable.

Model Layer (Data Layer)

The Model layer defines the MongoDB collections using Mongoose schemas. It is responsible for managing user information, uploaded study materials, and AI-generated learning resources while ensuring data validation and database interactions.

View Layer (API Response Layer)

Since AI StudyBuddy is a backend REST API, there is no traditional user interface in the View layer. Instead, the API returns structured JSON responses to the React frontend, allowing users to interact with the application through REST endpoints.

Controller Layer (Business Logic)

The Controller layer processes incoming API requests, validates user input, performs authentication and authorization, communicates with the database models, invokes the Google Gemini AI service when required, and returns the appropriate response to the client.

MVC Benefits

The MVC architecture separates data management, business logic, and client communication into independent components. This improves code readability, simplifies debugging and testing, and allows new features to be integrated without affecting existing modules.

Epic 2: Project Setup and Configuration

Creating the Project Folder

1. Create a new folder named **AI StudyBuddy** .
2. Open the project folder in **Visual Studio Code**.
3. Initialize the project using Node.js and install the required dependencies.

Project Initialization

Open the project folder in the VS Code terminal and execute:

```
npm init -y
```

Install the required dependencies:

```
npm install express mongoose bcryptjs jsonwebtoken cors dotenv multer @google/genai
```

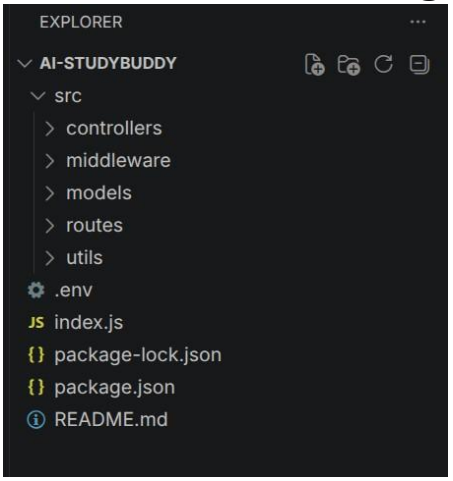
Install Nodemon for development:

```
npm install --save-dev nodemon
```

Create the Following Files

- index.js
- .env
- package.json
- .gitignore

Create the Following Folder Structure



Environment Configuration

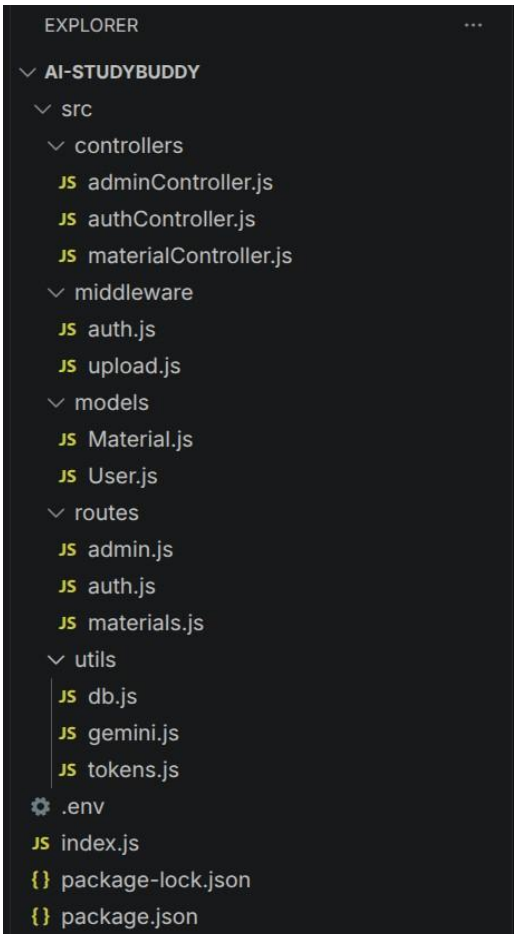
Create a **.env** file in the project root and configure the required environment variables.

PORT=5000
MONGO_URI=your_mongodb_connection_string
JWT_SECRET=your_jwt_secret
GEMINI_API_KEY=your_gemini_api_key

Epic 3: Backend Development

The AI StudyBuddy backend is developed using **Node.js**, **Express.js**, and **MongoDB**, following a modular architecture. The application separates routing, controllers, middleware, models, and utility modules to improve code maintainability and scalability. The backend exposes secure REST APIs for authentication, study material management, AI-powered content generation, and administrative operations.

Backend Structure



Core System Script Implementations

1. Application Entry Point (index.js)

The **index.js** file serves as the main entry point of the application. It initializes the Express server, loads environment variables, configures middleware, establishes the MongoDB connection, registers API routes, and starts the server after a successful database connection.

```
JS index.js > ...
1   require("dotenv").config();
2   require("express-async-errors");
3
4   const express = require("express");
5   const cors = require("cors");
6   const connectDB = require("./src/utils/db");
7
8   const app = express();
9
10  // Middleware
11  app.use(express.json());
12  app.use(cors({ origin: process.env.CLIENT_URL || "http://localhost:3000" }));
13
14  // Routes
15  app.use("/api/auth", require("./src/routes/auth"));
16  app.use("/api/materials", require("./src/routes/materials"));
17  app.use("/api/admin", require("./src/routes/admin"));
18
19  // Health check
20  app.get("/", (req, res) => res.json({ message: "AI StudyBuddy API is running" }));
21
22  // Global error handler
23  app.use((err, req, res, next) => {
24    console.error(err.message);
25    const status = err.status || 500;
26    res.status(status).json({ message: err.message || "Something went wrong" });
27  });
28
29  const PORT = process.env.PORT || 5000;
30
31  connectDB().then(() => {
32    app.listen(PORT, () => console.log(`Server running on port ${PORT}`));
33  });
34
```

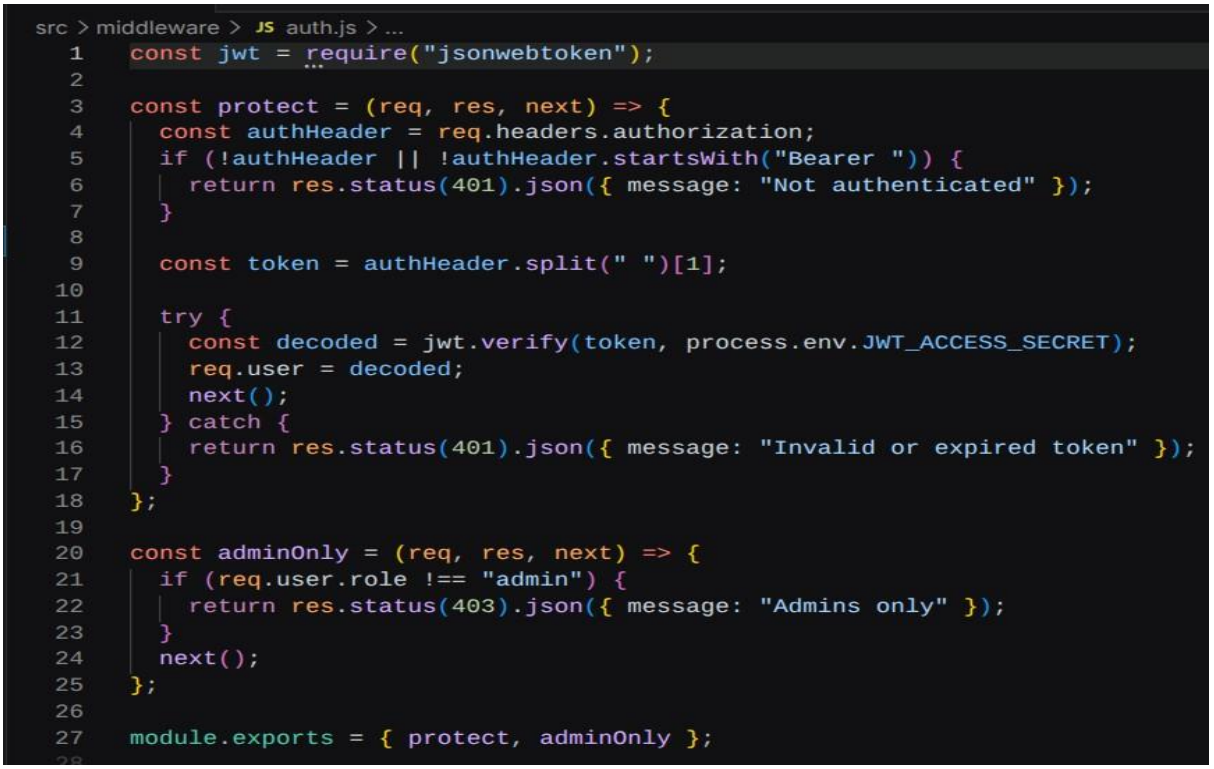
2. Database Connection (src/utils/db.js)

This module establishes a connection between the application and MongoDB using Mongoose. It ensures that the backend communicates with the database before processing any client requests.

```
src > utils > JS db.js > ...
1   const mongoose = require("mongoose");
2   const dotenv = require("dotenv");
3
4   dotenv.config();
5
6   const connectDB = async () => {
7     const conn = await mongoose.connect(process.env.MONGO_URI);
8     console.log(`MongoDB connected: ${conn.connection.host}`);
9   };
10
11  module.exports = connectDB;
12
```

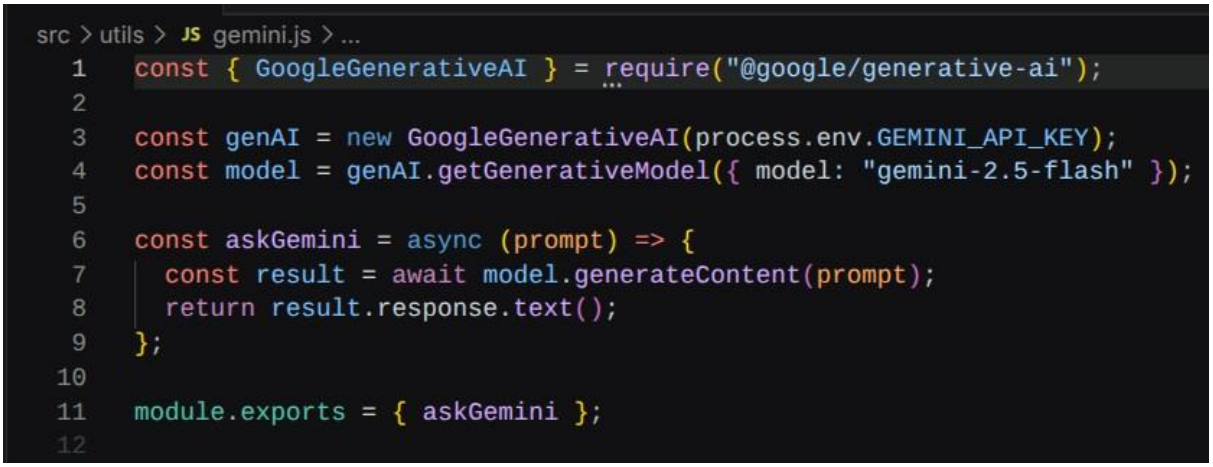
3. Authentication Middleware (src/middleware/auth.js)

The authentication middleware verifies JWT access tokens for protected routes. It authenticates users before granting access to secured APIs and includes role-based authorization to restrict administrative operations.



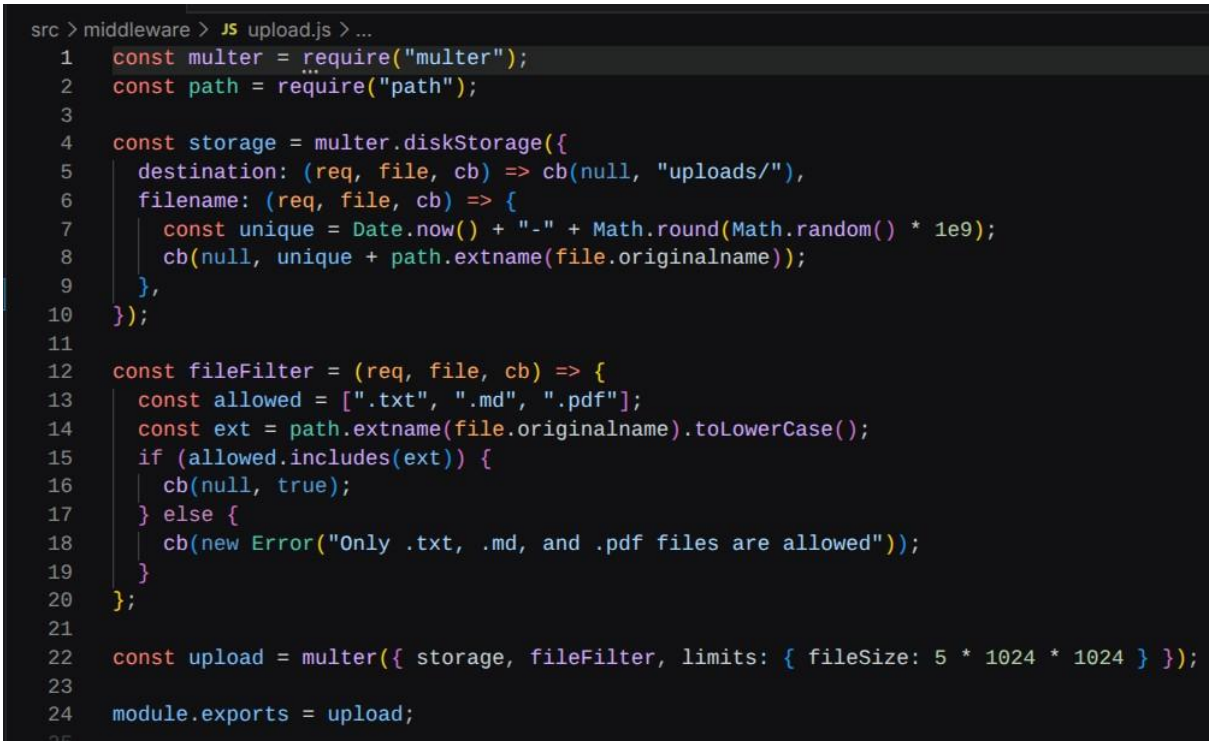
4. AI Integration (src/utils/gemini.js)

This utility integrates the backend with the **Google Gemini 2.5 Flash API**. It sends prompts to the AI model and returns generated responses for summaries, flashcards, quizzes, and study plans.



5. File Upload Middleware (src/middleware/upload.js)

The upload middleware handles study material uploads and validates incoming files before they are processed by the backend.



Epic 4: Database Configuration

The AI StudyBuddy backend uses **MongoDB** as its primary NoSQL database and **Mongoose ODM** for data modeling and schema validation. MongoDB stores user information, uploaded study materials, and AI-generated learning resources in a structured manner. Mongoose simplifies database operations by providing schemas, validation rules, and efficient query handling.

Configure MongoDB

Install Mongoose using the following command:

```
npm install mongoose
```

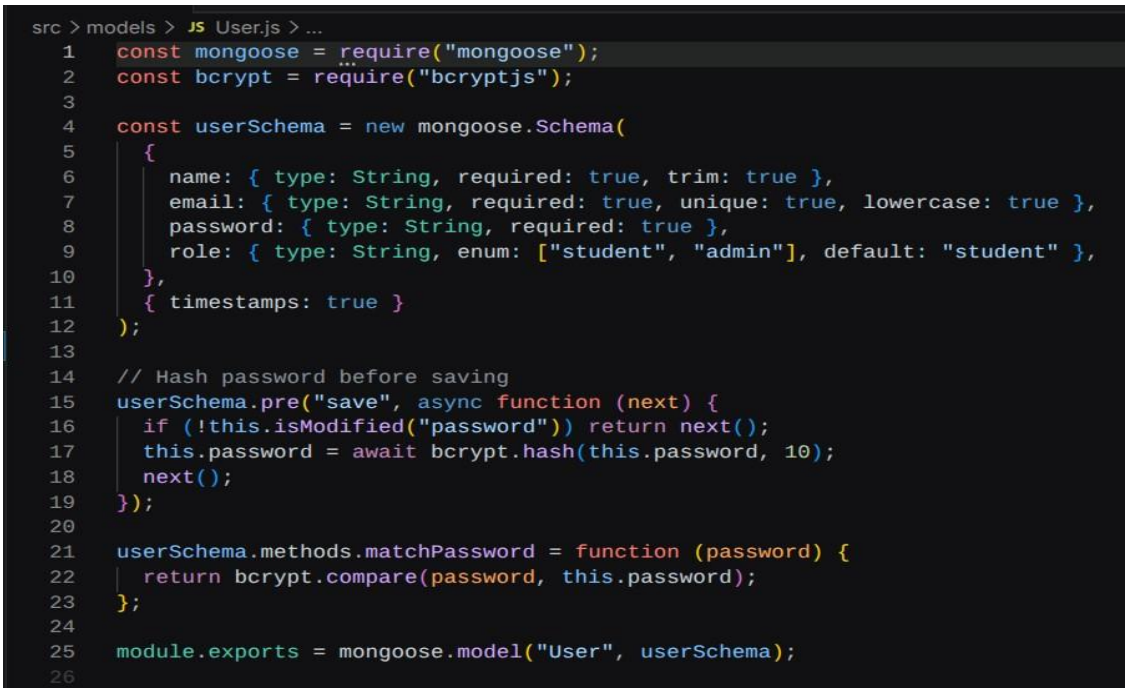
Create Database Connection

Before performing any database operations, establish a connection between the backend and MongoDB using Mongoose. The connection module is initialized during server startup to ensure reliable communication with the database.

Mongoose Schema Definitions

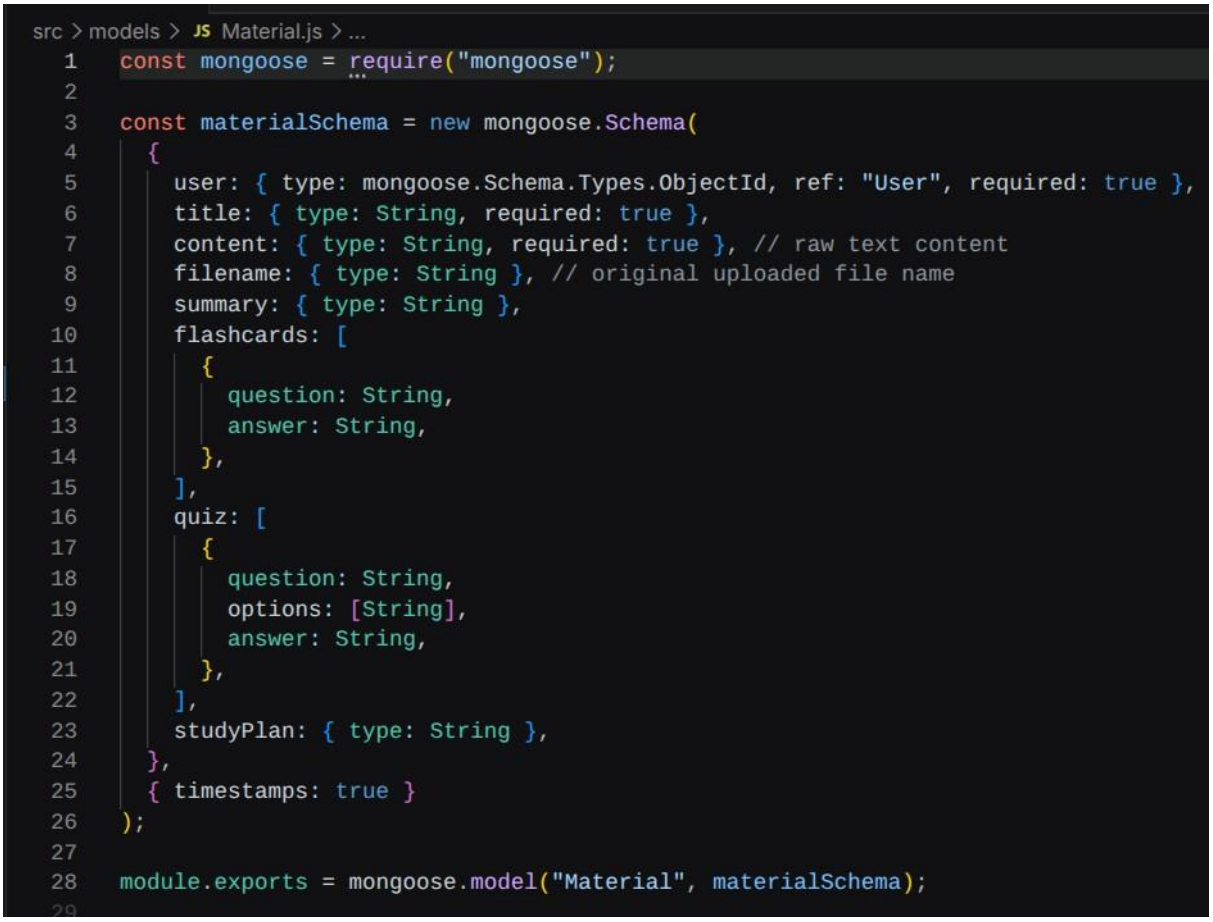
1. User Schema (models/User.js)

The **User** schema stores authentication and profile information for registered users. It includes fields such as name, email, password, and role. Passwords are securely encrypted before being stored in the database.



2. Study Material Schema (models/Material.js)

The **Study Material** schema stores the learning content uploaded by users. This content acts as the input for AI-powered features such as summaries, quizzes, flashcards, and study plans.



Database Collections

The AI StudyBuddy database consists of the following collections:

- **Users** – Stores user account and authentication details.
- **Study Materials** – Stores uploaded learning materials.

API Route Interface Endpoints

The backend exposes RESTful APIs to interact with the database and AI services.

1. Authentication APIs

Handles user registration, login, JWT authentication, and protected route access.

2. Study Material APIs

Allows users to upload, retrieve, update, and manage study materials.

3. AI Feature APIs

Provides endpoints for:

- AI Summary Generation
- AI Flashcard Generation
- AI Quiz Generation
- AI Study Plan Generation

4. Administrative APIs

Provides administrator-only endpoints for managing users and monitoring system activities.

Epic 5: Project Execution

The AI StudyBuddy backend can be executed locally using Node.js. Before running the application, ensure that MongoDB is running and the required environment variables are configured correctly. The backend exposes REST APIs that can be tested using Postman or Thunder Client.

Running the Backend Server

Install all required dependencies:

```
npm install
```

Start the development server:

```
npm start
```

After successful execution, the server connects to MongoDB and starts listening for incoming API requests.

API Testing

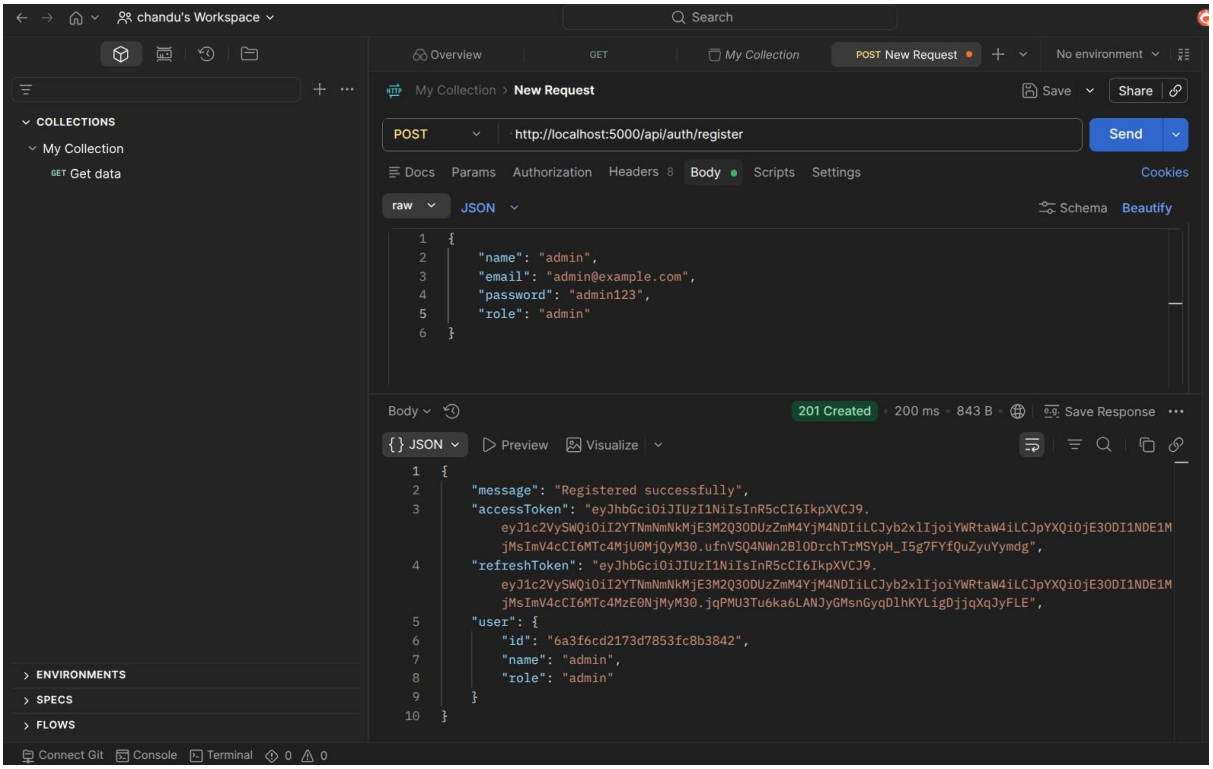
The backend APIs can be tested using **Postman**.

1. User Registration

Registers a new user by accepting the required details and securely storing the encrypted password in the MongoDB database.

Endpoint:

POST /api/auth/register

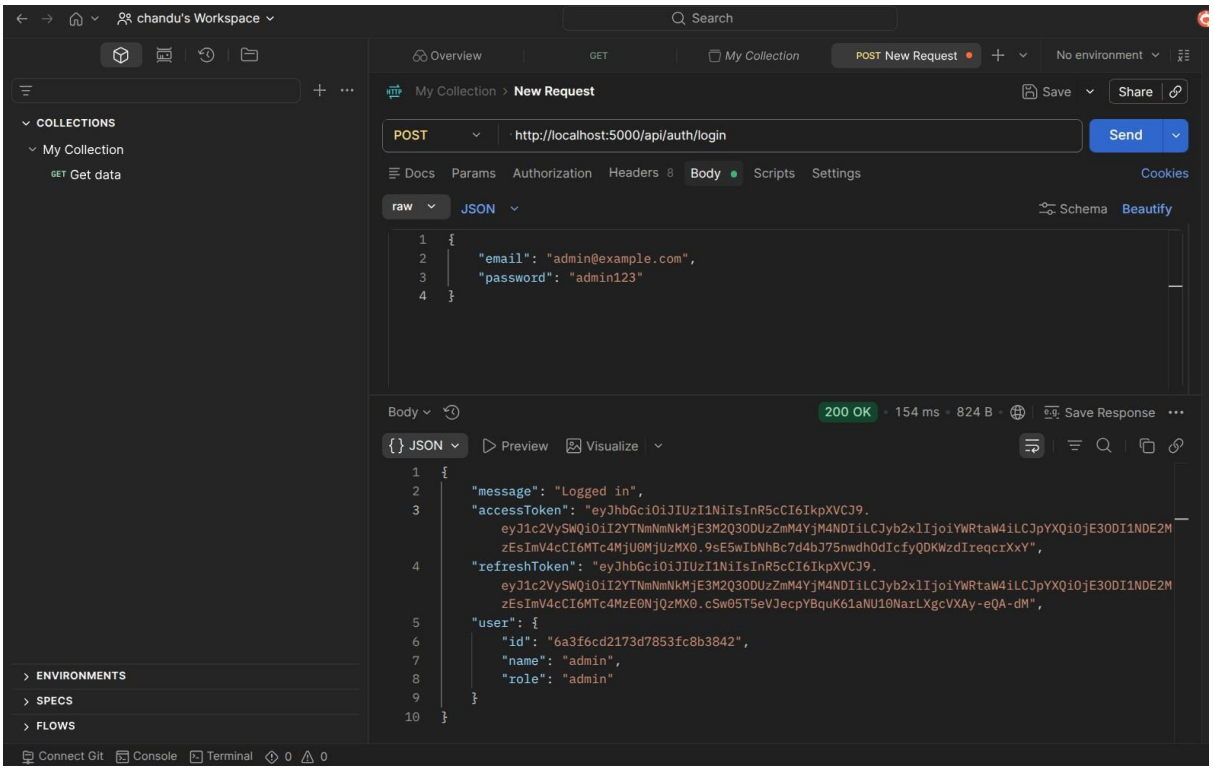


2. User Login

Authenticates the registered user and generates a JWT access token for accessing protected APIs.

Endpoint:

POST /api/auth/login

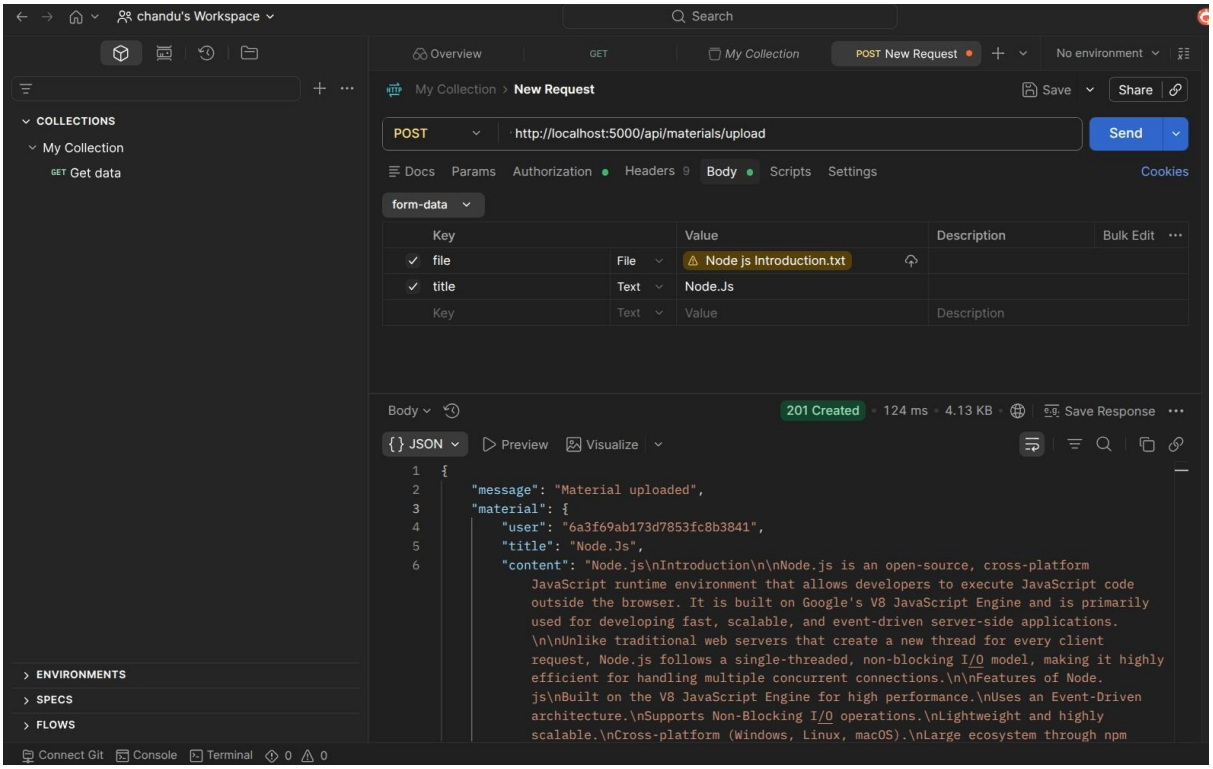


3. Upload Study Material

Allows authenticated users to upload study material that will be used as input for AI-powered content generation.

Endpoint:

POST /api/material/upload

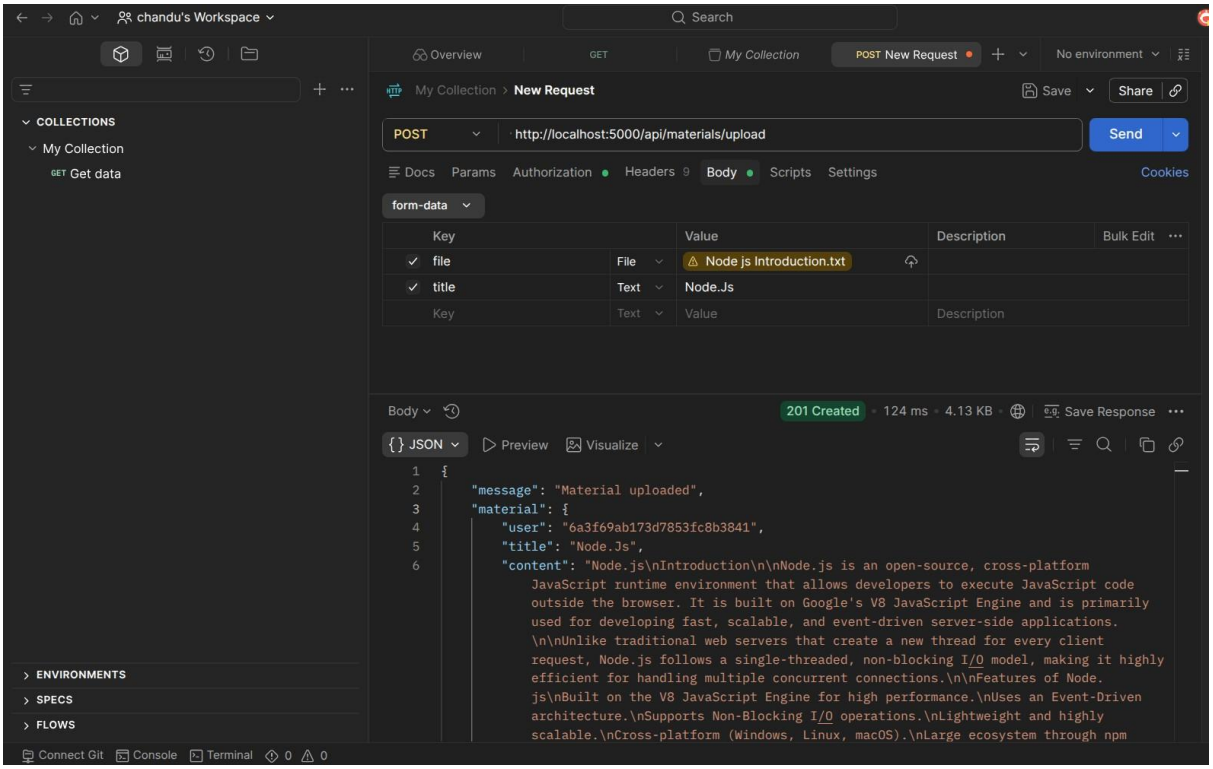


4. Generate AI Summary

Sends the uploaded study material to the Google Gemini API and generates a concise summary.

Endpoint:

POST /api/materials/:id/summarize

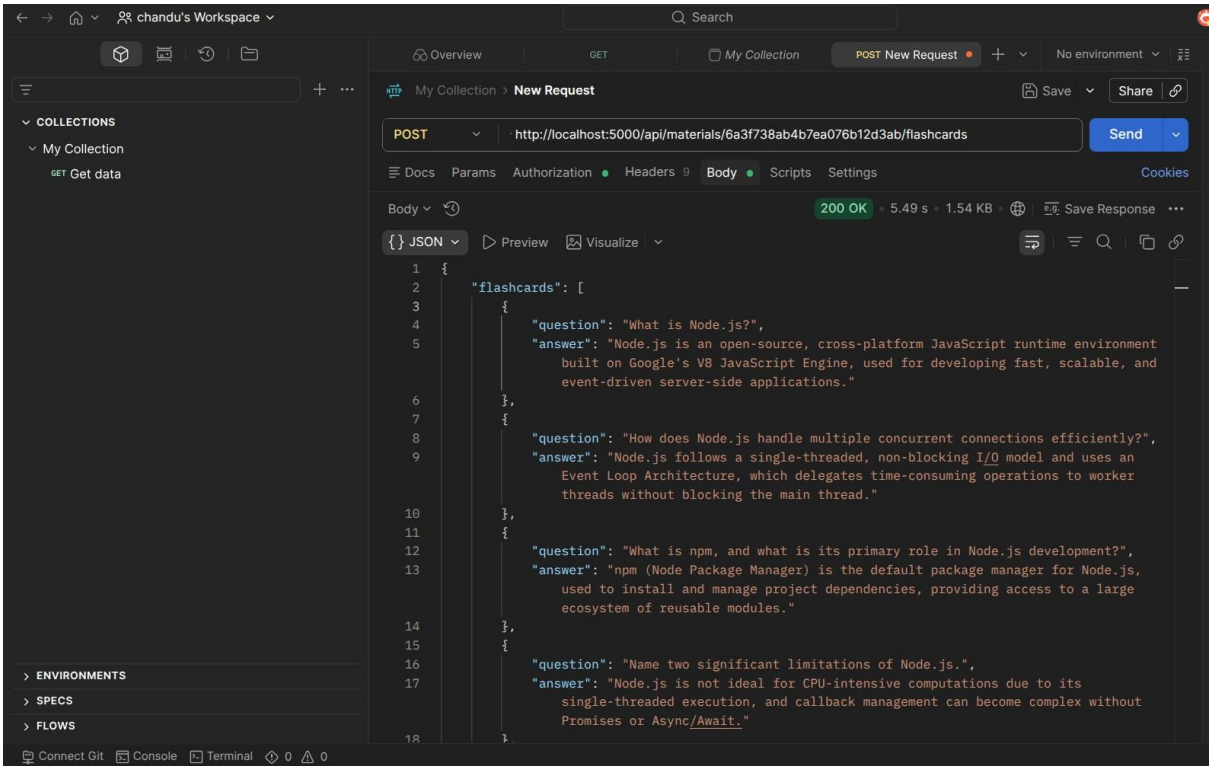


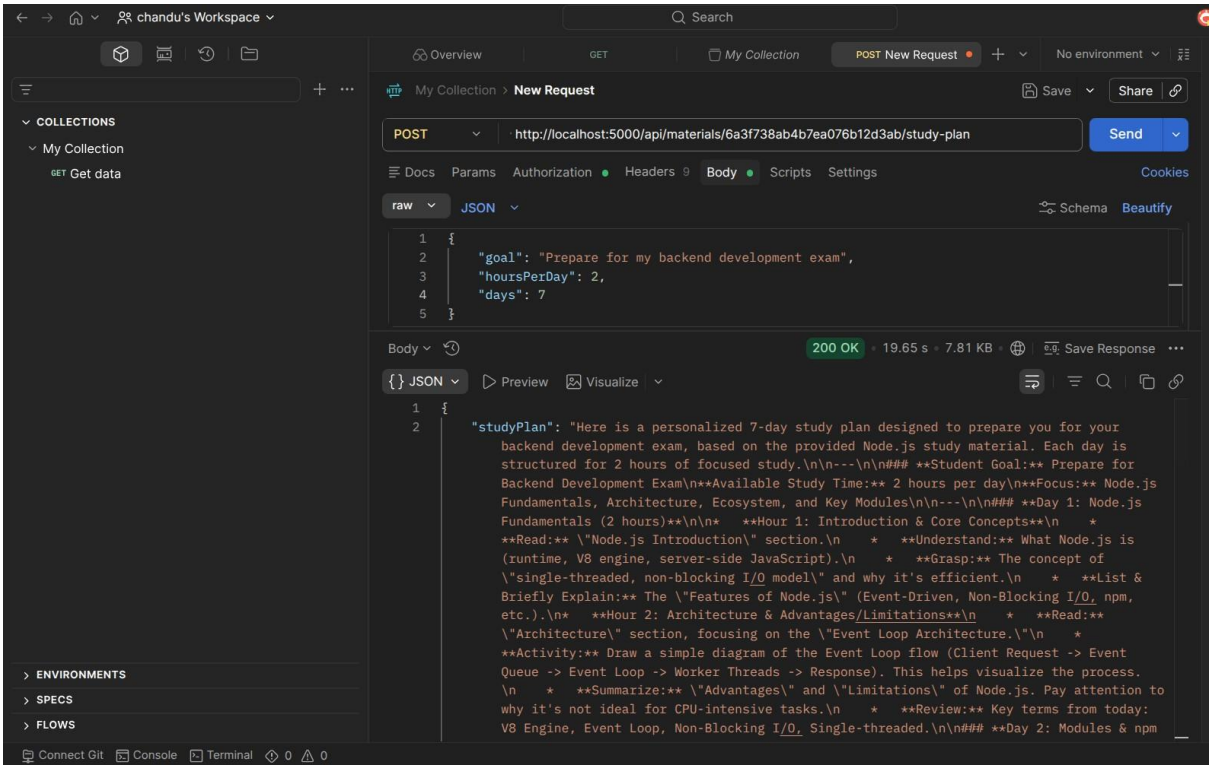
5. Generate AI Flashcards

Creates AI-generated flashcards from the uploaded study material for effective revision.

Endpoint:

POST /api/ai/flashcards





DEMO VIDEO LINK : https://drive.google.com/file/d/16-Nol6etUbwVo8PnxcoSIQF-ZqvKd70s/view?usp=sharing

GIT HUB LINK: https://github.com/SHERAFINCATHERINAN/AI-Studybuddy.git

CODE DRIVE LINK:
https://drive.google.com/drive/folders/1mfJrGr63VDaF4zhXO7j9DIS_EWGJWjK0